Telecom churn prediction

Introduction

- In the fiercely competitive telecom industry, customer loyalty is more fragile than ever.
- Studies reveal that acquiring a new customer can cost up to **five times more** than retaining an existing one.
- Yet, many companies remain reactive, addressing churn only after it occurs.
- Imagine the advantage if we could **predict** which customers are likely to leave and take proactive steps to retain them.
- This is where **Machine Learning** becomes a game-changer, transforming raw data into actionable insights to **reduce churn** and **enhance customer satisfaction**.

Understanding Customer Churn in Telecom

- **Customer churn** means when a customer decides to stop using a company's service.
- In the **telecom industry**, understanding churn is critical due to high competition and customer acquisition costs.
- Using **Machine Learning**, we can predict which customers are likely to churn — so companies can take action before it's too late!

Key features in our telecom dataset

Feature	Description
customer_id	Unique identifier for each customer.
telecom_partner	The telecom partner associated with the customer.
gender	The gender of the customer.
age	The age of the customer
state	The Indian state in which the customer is located.
city	The city in which the customer is located.
pincode	The pincode of the customer's location.
date_of_registration	The date on which the customer registered with the telecom partner.
num_dependents	The number of dependents (e.g. children) the customer has.

Key features in our telecom dataset

Feature	Description
estimated_salary	The customer's estimated salary.
calls_made	The number of calls made by the customer
sms_sent	The number of SMS messages sent by the customer.
data_used	The amount of data used by the customer
churn	Binary variable indicating whether the customer has churned or not (1 = churned, 0 = not churned).

Data preprocessing steps

Handling negative values

- Detected and converted all negative values to positive.
- Ensured numerical features are clean and logical.

Converting Date Column

- Extracted year, month, and day for better time-based analysis.
- Transformed date_of_registration into DateTime format.

Detecting & Handling Outliers

- Conducted an outlier analysis using statistical methods.
- Ensured that extreme values do not skew the model's performance.

Final Check

- Confirmed that the dataset is now clean, structured, and ready for EDA and modeling

Exploratory data analysis (EDA)

Churn Distribution:

- Identified imbalance between churned and non-churned customers.

Correlation Analysis:

- Used a heatmap to explore relationships between numerical features.

Churn by Telecom Partner:

- Compared churn rates across telecom providers; all had similar churn levels.

Churn by Gender:

- Analyzed gender-wise churn distribution — males showed slightly higher churn.

Usage Behavior:

- Compared number of calls and SMS sent between churned and retained users.
- Visualized patterns using histograms and scatterplots.
- Found that churned users made fewer calls.

Exploratory data analysis (EDA)

Data Usage vs. Salary, Age, Gender:

- Explored relationships between data usage and features like salary and age.
- Younger users and those with higher income showed varying usage patterns.

Regional Usage Patterns:

- Top states and cities were ranked by average data usage.
- User distribution was analyzed by state.

Feature Engineering Insights:

- Created new features like calls_per_sms, data_per_call, and data_per_dependent for deeper behavioral understanding.

Model building

Accuracies for churn prediction

Random forest	79%
XGBoost	79%
XGBoost after (Hyperparameter tuning)	79.95%
Voting classifier	79.95%
Staking classifier	79.95%
Random forest with SMOTE	87%

SUMMARY & CONCLUSION

- The project aimed to develop a model to predict customer churn for telecom companies.
- After using SMOTE, the highest achieved accuracy was 87%.
- This helps telecom companies understand their customers better and improve retention.
- In conclusion, our approach offers valuable insights and opportunities for enhancing customer loyalty in the telecom sector.

THANK YOU